

Market Integration, Platformization, and Local Multipliers: The Spatial Reorganization of Value Capture

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July 16, 2026

Abstract

We study how market integration and platformization reorganize where transaction value is captured. In a small open region, integration lowers the platform channel's price and may expand autonomous local income, but shifts spending toward an intermediary whose local-income share depends on local organizational capacity. We solve the income fixed point. Closed-form thresholds show when access gains dominate capture losses and when consumer gains coexist with falling local income. Stronger local organization raises effective multipliers and can reverse adverse integration effects. When the platform retains less value locally, capture drag peaks before platform penetration reaches one-half.

Keywords: market integration; digital platforms; local multipliers; value capture; spatial incidence

JEL codes: F15; L81; R11; R12

1 Introduction

Market integration expands access to products and markets, but digital platforms also reorganize the intermediaries through which transactions are matched, settled, and fulfilled. Expenditure in a region need not become income there: platform fees, seller profits, and service revenue may accrue elsewhere. The local effect therefore depends on both market-access gains and the share of transaction value captured locally. Because captured income finances demand for local services, this spatial incidence also changes the effective local multiplier.

Evidence from rural China shows that e-commerce access can lower living costs without generating comparable gains for local producers and workers (Couture et al., 2021). Online trade changes regional consumption and real wages (Fan et al., 2018), and retail entry can generate price gains alongside local labor and profit effects (Atkin et al., 2018).

The local-multiplier literature studies how tradable-employment or fiscal-spending shocks propagate across local activity (Moretti, 2010; Chodorow-Reich, 2019). Retail channel organization, however, is not the object of analysis in these studies. With platforms, otherwise identical expenditure flows need not generate the same local income: seller margins, fulfillment revenue, platform payments, and ownership returns may accrue outside the region. Local propagation therefore depends on both responding and the transaction value retained in each round.

We introduce this channel-specific retention margin into a small-open-region multiplier framework. Integration lowers the relative platform price and may expand autonomous base income, while endogenous channel substitution changes the average share of expenditure captured as local labor, service, and locally owned business income. Organizational capacity κ shifts platform retention between lower and upper bounds; it does not require local platform ownership. The model yields

closed-form access-capture thresholds for nominal and consumption-equivalent real income and ordered capacity thresholds: weak local organization can make capture losses dominate, intermediate capacity can preserve consumer gains despite falling local income, and strong capacity can make access gains dominate while raising the effective multiplier. For a fixed capacity level with $\Delta(\kappa) > 0$, marginal capture drag is strictly hump-shaped and peaks before the platform share reaches one-half. This is a short-run local equilibrium, not a model of platform pricing or a multi-region quantitative spatial equilibrium.

2 Model

Consider a small region L embedded in an outside economy H . Channel P is an outside digital-platform channel whose relative price responds to market integration; A is a local-agent channel.

Assumption 1 (Primitives). Let $z \in \mathbb{R}$ index the degree of market integration; a higher z denotes stronger integration and a lower platform-channel trade cost. $\alpha, \beta > 0$, $\alpha + \beta < 1$, $a \in (0, 1)$, $\eta > 1$, and

$$p_A, \bar{p}_P, \bar{B} > 0, \quad G, \varepsilon \geq 0, \quad \rho_A \in [0, 1], \quad \kappa \in [0, 1], \\ 0 \leq \underline{\rho}_P < \bar{\rho}_P \leq 1, \quad \delta_\rho \equiv \bar{\rho}_P - \underline{\rho}_P > 0.$$

The channel- P price is $p_P(z) = \bar{p}_P e^{-z}$, and autonomous base income is $B(z) = \bar{B} e^{\varepsilon z}$.

Assumption 2 (Local-income closure). The outside-good price and local wage are one. A competitive constant-returns sector produces the nontradable with $N = L_N$, so its price is $p_N = 1$; labor or capacity is elastically supplied at its fixed opportunity cost. Each dollar spent through channel c generates ρ_c dollars of local labor, service, and locally owned business income; the remainder accrues to H . Local organizational capacity κ changes only the spatial incidence of platform transactions, not prices or preferences. Autonomous base income B is local value added outside the modeled retail and nontradable sectors and excludes income generated by G . Residents own all local income and share the preferences below. Finally, G is retail expenditure funded from outside L and allocated across channels by $s(z)$. Each induced spending round purchases newly produced output and creates income according to these incidence shares.

We treat L as small and take the platform-price schedule $p_P(z)$ as exogenous to the region: integration moves the price along this schedule, whereas local income and expenditure do not feed back into it. We take the outside good as numeraire and assume that the fixed opportunity cost of elastically supplied local labor equals one in those units. Because the competitive nontradable technology uses only local labor, $N = L_N$, zero profit implies $p_N = w = 1$, so all nontradable revenue is local labor income. These restrictions isolate channel-specific transaction incidence as the model's spatial-capture margin; ρ_c is therefore a total local-income share, not necessarily a labor-cost share.

A representative resident solves

$$\max_{M, N, Z \geq 0} M^\alpha N^\beta Z^{1-\alpha-\beta} \quad \text{s.t.} \quad P_M M + N + Z = Y. \quad (1)$$

Retail consumption combines a local-agent channel A and platform channel P :

$$M = \left[(1-a)^{1/\eta} M_A^{(\eta-1)/\eta} + a^{1/\eta} M_P^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)}. \quad (2)$$

Conditional on M , the household minimizes

$$\min_{M_A, M_P \geq 0} \{p_A M_A + p_P(z) M_P\} \quad \text{s.t.} \quad (2). \quad (3)$$

Lemma 1 (Household and channel demand). *For $Y > 0$, both problems have unique interior solutions. Expenditure on (M, N, Z) is $(\alpha Y, \beta Y, (1 - \alpha - \beta)Y)$. Moreover,*

$$P_M(z) = \left[(1 - a)p_A^{1-\eta} + ap_P(z)^{1-\eta} \right]^{1/(1-\eta)}, \quad (4)$$

$$M_c = a_c \left(\frac{p_c}{P_M} \right)^{-\eta} M, \quad (a_A, a_P) = (1 - a, a), \quad (5)$$

$$s(z) = \frac{ap_P(z)^{1-\eta}}{(1 - a)p_A^{1-\eta} + ap_P(z)^{1-\eta}} \in (0, 1). \quad (6)$$

For every finite z ,

$$\frac{d \ln P_M}{dz} = -s, \quad \frac{ds}{dz} = (\eta - 1)s(1 - s) > 0. \quad (7)$$

Platform transactions retain the share

$$\rho_P(\kappa) = \underline{\rho}_P + \kappa \delta_\rho, \quad (8)$$

where κ summarizes the local seller, fulfillment, and service organization able to participate in platform trade. Average local value capture is

$$\omega(z, \kappa) = (1 - s)\rho_A + s\rho_P(\kappa), \quad (9)$$

$$\omega_z = -\Delta(\kappa)(\eta - 1)s(1 - s), \quad \omega_\kappa = s\delta_\rho > 0, \quad (10)$$

$$\Delta(\kappa) \equiv \rho_A - \rho_P(\kappa). \quad (11)$$

The benchmark condition $\Delta(\kappa) > 0$ is imposed only where stated; a lower platform retention share is not part of the definition of a platform.

Residents spend αY through the retail channels and βY on the local nontradable. With $X_A = (1 - s)(\alpha Y + G)$ and $X_P = s(\alpha Y + G)$, Assumption 2 gives nontradable revenue $p_N N = N = \beta Y$ and channel income $\rho_A X_A + \rho_P(\kappa) X_P = \omega(\alpha Y + G)$.

Definition 1 (Local-income equilibrium). For given z, κ , and G , an equilibrium is

$$(M_A, M_P, M, N, Z, s, \omega, Y, R)$$

that satisfies household optimization, channel expenditure minimization, (9), and

$$Y = B(z) + \beta Y + \omega(z, \kappa)[\alpha Y + G], \quad (12)$$

with R the consumption-equivalent real-income index implied by the household problem. The objects P_M and D are derived uniquely from z and the primitives.

Lemma 2 (Existence and uniqueness). *An equilibrium exists, is unique, and has $Y > 0$. For fixed z , the income map*

$$T(Y) = B + \omega G + (\beta + \alpha \omega)Y$$

is a contraction on $\mathbb{R}_+$; iteration from any $Y_0 \geq 0$ converges to

$$Y(z, \kappa, G) = \frac{B(z) + \omega(z, \kappa)G}{D(z, \kappa)}, \quad D(z, \kappa) = 1 - \beta - \alpha \omega(z, \kappa) > 0. \quad (13)$$

Indeed, $0 \leq \beta + \alpha \omega \leq \alpha + \beta < 1$. Household optimization then gives consumption-equivalent real income

$$R(z, \kappa, G) = K \frac{Y(z, \kappa, G)}{P_M(z)^\alpha}, \quad K = \alpha^\alpha \beta^\beta (1 - \alpha - \beta)^{1-\alpha-\beta}. \quad (14)$$

This is an indirect consumption index conditional on Assumption 2, not social welfare including labor supply and opportunity costs.

3 Results

Proposition 1 (Local-income and gross-spending multipliers). *Holding z , κ , and G fixed, treat realized B as an independent autonomous-income injection (equivalently, vary its level shifter $\bar{B}$).*

$$m_B(\omega) \equiv \frac{\partial Y}{\partial B} = \frac{1}{1 - \beta - \alpha\omega}, \quad (15)$$

$$\mathcal{M}_G(\omega) \equiv \frac{\partial Y}{\partial G} = \frac{\omega}{1 - \beta - \alpha\omega} = \omega m_B(\omega) = \omega \sum_{k=0}^{\infty} (\beta + \alpha\omega)^k. \quad (16)$$

Let $m_0 = (1 - \alpha - \beta)^{-1}$. Then $\mathcal{M}_G(\omega) \leq m_0\omega$, with equality exactly at $\omega \in \{0, 1\}$ and strict inequality for $0 < \omega < 1$. For $\omega > 0$,

$$\frac{\partial \ln \mathcal{M}_G}{\partial \ln \omega} = \frac{1 - \beta}{1 - \beta - \alpha\omega} > 1. \quad (17)$$

The multiplier m_B propagates one dollar already formed as local income. The gross-spending multiplier $\mathcal{M}_G$ additionally applies first-round incidence ω . Its series shows why $m_0\omega$ overstates propagation: the latter applies leakage only initially, whereas retail leakage recurs when induced income finances new retail expenditure.

Set $G = 0$, take $z_1 > z_0$, let $B_j = B(z_j)$, $D_j = D(z_j, \kappa)$, and $P_{Mj} = P_M(z_j)$ for $j \in \{0, 1\}$, and define

$$\mathcal{B} = \frac{B_1}{B_0}, \quad \mathcal{D} = \frac{D_1}{D_0}, \quad \mathcal{P} = \frac{P_{M1}}{P_{M0}}. \quad (18)$$

Proposition 2 (Finite access-capture thresholds). *Fix κ . Suppose $G = 0$ and $\Delta(\kappa) > 0$. Then $\mathcal{D} > 1$, $0 < \mathcal{P} < 1$, and*

$$\frac{Y_1}{Y_0} = \frac{\mathcal{B}}{\mathcal{D}}, \quad \frac{R_1}{R_0} = \frac{\mathcal{B}}{\mathcal{D}\mathcal{P}^\alpha}. \quad (19)$$

The three strict regions are: (i) both incomes rise if $\mathcal{B} > \mathcal{D}$; (ii) nominal income falls but real income rises if $\mathcal{D}\mathcal{P}^\alpha < \mathcal{B} < \mathcal{D}$; and (iii) both fall if $\mathcal{B} < \mathcal{D}\mathcal{P}^\alpha$. At $\mathcal{B} = \mathcal{D}$, nominal income is unchanged and real income rises; at $\mathcal{B} = \mathcal{D}\mathcal{P}^\alpha$, nominal income falls and real income is unchanged. These cases are exhaustive; nominal income cannot rise while real income falls.

Two consequences are useful. Since $\mathcal{B} \geq 1$, holding $\mathcal{D}$ and $\mathcal{P}$ fixed while allowing $\varepsilon \geq 0$, the joint-loss region is nonempty if and only if $\mathcal{D}\mathcal{P}^\alpha > 1$. At $\varepsilon = 0$, the local condition $\Delta(\kappa)(\eta - 1)(1 - s) > D(s, \kappa)$ makes both marginal effects negative.

For marginal changes at $G = 0$, define

$$\Lambda(s, \kappa) = \frac{\alpha\Delta(\kappa)(\eta - 1)s(1 - s)}{D(s, \kappa)}. \quad (20)$$

Corollary 1 (Organizational capacity and integration). *Suppose $G = 0$. Then*

$$\frac{d \ln Y}{dz} = \varepsilon - \Lambda(s, \kappa), \quad \frac{d \ln R}{dz} = \varepsilon + \alpha s - \Lambda(s, \kappa). \quad (21)$$

Moreover,

$$\frac{\partial \ln Y}{\partial \kappa} = \frac{\alpha s \delta_\rho}{D} > 0, \quad (22)$$

$$\frac{\partial \Lambda}{\partial \kappa} = -\frac{\alpha \delta_\rho (\eta - 1)s(1 - s)(1 - \beta - \alpha \rho_A)}{D^2} < 0, \quad (23)$$

$$\frac{\partial \ln \mathcal{M}_G}{\partial \kappa} = \frac{(1 - \beta)s \delta_\rho}{\omega D} > 0 \quad \text{if } \omega > 0. \quad (24)$$

Let $h(s) = (\eta - 1)(1 - s)$ and $d_A = 1 - \beta - \alpha\rho_A$. For $q \geq 0$ satisfying $\Lambda(s, 0) > q > \Lambda(s, 1)$, the condition implies $q < h(s)$; define

$$\kappa(q) = \frac{1}{\delta_\rho} \left[\rho_A - \underline{\rho}_P - \frac{q d_A}{\alpha s [h(s) - q]} \right]. \quad (25)$$

This is the unique capacity in $(0, 1)$ at which $\Lambda(s, \kappa) = q$. If $\Lambda(s, 0) > \varepsilon + \alpha s > \varepsilon > \Lambda(s, 1)$, then

$$0 < \kappa_R^* \equiv \kappa(\varepsilon + \alpha s) < \kappa_Y^* \equiv \kappa(\varepsilon) < 1.$$

Both marginal effects are negative below κ_R^* ; only real income rises between the two thresholds; and both rise above κ_Y^* .

The two access gains are ε for nominal income and $\varepsilon + \alpha s$ after adding the consumer-price effect. Organizational capacity lowers the opposing capture drag and increases the multiplier. If $\rho_P(\kappa) \geq \rho_A$, then $\Lambda \leq 0$, so integration cannot reduce either income measure under $\varepsilon \geq 0$.

Corollary 2 (Intermediate-penetration peak). *Fix κ with $\Delta(\kappa) > 0$, and let $d_A = 1 - \beta - \alpha\rho_A$. The function $s \mapsto \Lambda(s, \kappa)$, extended continuously to $[0, 1]$, is strictly concave and has the unique maximizer*

$$s^\dagger(\kappa) = \frac{1}{1 + \sqrt{1 + \alpha\Delta(\kappa)/d_A}} < \frac{1}{2}. \quad (26)$$

Writing $t(\kappa) = \sqrt{1 + \alpha\Delta(\kappa)/d_A}$, its maximum is

$$\Lambda_{\max}(\kappa) = (\eta - 1) \frac{t(\kappa) - 1}{t(\kappa) + 1}. \quad (27)$$

Over interior capacity levels for which $\Delta(\kappa) > 0$, $\partial s^\dagger / \partial \kappa > 0$ and $\partial \Lambda_{\max} / \partial \kappa < 0$; the corresponding feasible one-sided derivatives apply at $\kappa \in \{0, 1\}$.

The peak is a marginal, not cumulative, effect. Platform substitution is weak near either channel-share boundary, while stronger local organization reduces the retention gap at every interior share. Figure 1 illustrates both implications with an admissible parameterization satisfying the endpoint-crossing condition; it is not a calibration.

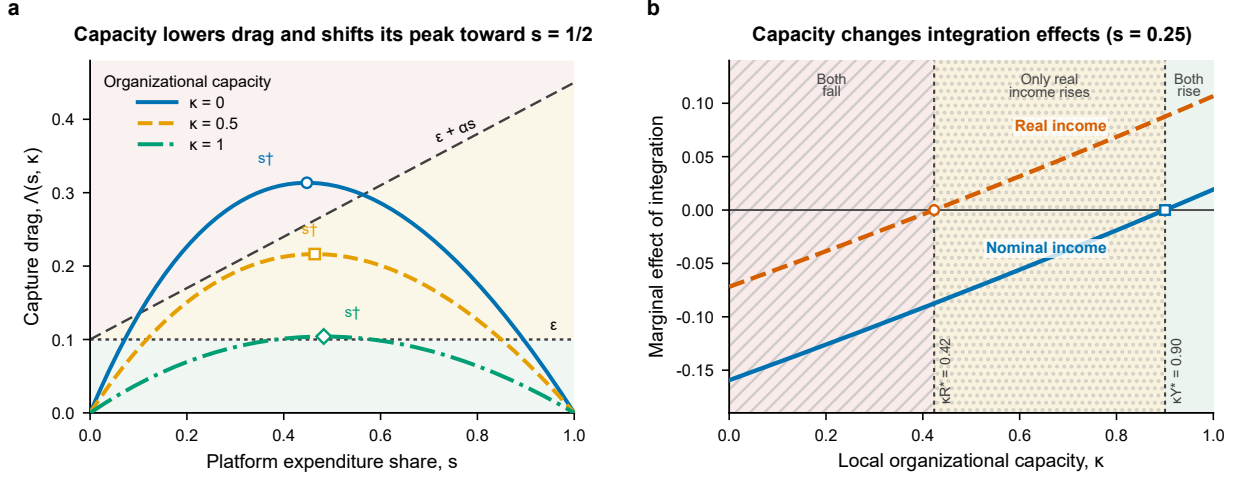


Figure 1: Value-capture mechanisms. (a) At fixed low, medium, and high organizational capacity, capture drag is hump-shaped in platform penetration; stronger organization lowers the peak and moves it toward one-half. The horizontal ε line and upward-sloping $\varepsilon + \alpha s$ line delimit the three sign regions. (b) At a fixed platform share, the nominal- and real-income effects of integration increase with organizational capacity and cross zero at $\kappa_R^* < \kappa_Y^*$, producing the three sign regions in Corollary 1. Parameters are $\alpha = 0.35$, $\beta = 0.25$, $\eta = 4$, $\rho_A = 0.8$, $\underline{\rho}_P = 0.1$, $\bar{\rho}_P = 0.6$, and $\varepsilon = 0.1$; panel (b) fixes $s = 0.25$.

4 Conclusion

Market integration can expand access while platformization relocates the income generated by transactions. The model separates these forces and shows how the retained share governs both first-round income formation and subsequent local demand. Low organizational capacity can make capture losses dominate; intermediate capacity can preserve consumer gains despite lower local income; and sufficiently high capacity can make access gains dominate while raising the effective multiplier. When $\Delta(\kappa) > 0$, the marginal capture drag is greatest during intermediate platform penetration, before either channel dominates. The framework is deliberately a short-run local equilibrium: it takes the platform-price schedule and production locations as given, abstracts from tax incidence, and evaluates consumption-equivalent real income rather than full social welfare. These restrictions isolate channel-specific retention as the link between platform intermediation and local income formation; embedding this mechanism in a multi-region equilibrium is a natural extension.

Data availability

No data were used for the research described in this article.

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A Proofs and Boundary Cases

Proof of Lemma 1. Write $\gamma = 1 - \alpha - \beta > 0$. Maximizing log utility in (1) gives

$$\frac{\alpha}{M} = \lambda P_M, \quad \frac{\beta}{N} = \lambda, \quad \frac{\gamma}{Z} = \lambda. \quad (\text{A.1})$$

The budget constraint implies $\lambda = 1/Y$, hence

$$M = \frac{\alpha Y}{P_M}, \quad N = \beta Y, \quad Z = \gamma Y. \quad (\text{A.2})$$

Strictly positive shares, prices, and income make this solution interior. Substitution yields

$$U = \alpha^\alpha \beta^\beta \gamma^\gamma \frac{Y}{P_M^\alpha}, \quad (\text{A.3})$$

which establishes (14).

For the inner problem, let $\mathcal{C}(M_A, M_P)$ denote the right side of (2). A corner cannot minimize cost. For example, as $M_A \downarrow 0$ with $M_P > 0$, $\mathcal{C}_A/\mathcal{C}_P \rightarrow \infty$; moving along the isoquant has marginal cost $p_A - p_P \mathcal{C}_A/\mathcal{C}_P < 0$ near the boundary. The symmetric argument excludes $M_P = 0$. We may therefore use the interior Lagrangian

$$\mathcal{L} = p_A M_A + p_P M_P + \mu [M - \mathcal{C}(M_A, M_P)]. \quad (\text{A.4})$$

Its first-order conditions imply

$$\frac{M_A}{M_P} = \frac{1-a}{a} \left(\frac{p_A}{p_P} \right)^{-\eta}. \quad (\text{A.5})$$

Using the binding constraint gives

$$M_A = (1-a) \left(\frac{p_A}{P_M} \right)^{-\eta} M, \quad M_P = a \left(\frac{p_P}{P_M} \right)^{-\eta} M, \quad (\text{A.6})$$

and unit cost is (4). Strict quasiconcavity of the CES aggregator makes its expenditure-requirement set strictly convex, so the positive-price solution is unique. All terms in (6) are positive, hence $0 < s < 1$ for finite z .

Let $Q = (1-a)p_A^{1-\eta} + ap_P^{1-\eta}$. Since $d \ln p_P / dz = -1$,

$$\frac{d \ln P_M}{dz} = \frac{ap_P^{1-\eta}}{Q} \frac{d \ln p_P}{dz} = -s. \quad (\text{A.7})$$

Also,

$$\frac{s}{1-s} = \frac{a}{1-a} \left(\frac{p_P}{p_A} \right)^{1-\eta} \implies s' = (\eta-1)s(1-s). \quad (\text{A.8})$$

Differentiating (8) and (9) gives (10). $\square$

Proof of Lemma 2. For fixed z , T maps $\mathbb{R}_+$ into itself because $B + \omega G > 0$. Since $s \in (0, 1)$ and both channel-retention shares lie in $[0, 1]$, $0 \leq \omega \leq 1$. For any $Y, \tilde{Y} \geq 0$,

$$|T(Y) - T(\tilde{Y})| = (\beta + \alpha\omega)|Y - \tilde{Y}|, \quad 0 \leq \beta + \alpha\omega \leq \alpha + \beta < 1. \quad (\text{A.9})$$

The contraction theorem gives a unique fixed point and global convergence of $Y_{n+1} = T(Y_n)$. Solving the fixed-point equation gives (13); its numerator and denominator are positive. Lemma 1 then uniquely determines all remaining objects. $\square$

Proof of Proposition 1. Fixing z , κ , and G , treat realized B as an independent autonomous-income injection, equivalently a change in the level shifter $\bar{B}$. Differentiating (13) in that injection and in gross external retail spending G gives (15) and (16); the series converges because $\beta + \alpha\omega < 1$. Moreover,

$$m_0\omega - \mathcal{M}_G(\omega) = \frac{\alpha\omega(1-\omega)}{(1-\alpha-\beta)(1-\beta-\alpha\omega)}. \quad (\text{A.10})$$

Every denominator is positive, proving the inequality and both equality cases. Direct differentiation gives (17) and

$$\mathcal{M}_G''(\omega) = \frac{2\alpha(1-\beta)}{(1-\beta-\alpha\omega)^3} > 0. \quad (\text{A.11})$$

□

Proof of Proposition 2. At fixed κ , $\Delta(\kappa) > 0$ and Lemma 1 imply $\omega_1 < \omega_0$, $D_1 > D_0$, and $P_{M1} < P_{M0}$. With $G = 0$,

$$\frac{Y_1}{Y_0} = \frac{B_1/D_1}{B_0/D_0} = \frac{\mathcal{B}}{\mathcal{D}}, \quad \frac{R_1}{R_0} = \frac{\mathcal{B}}{\mathcal{D}\mathcal{P}^\alpha}. \quad (\text{A.12})$$

Because $0 < \mathcal{P}^\alpha < 1$, $\mathcal{D}\mathcal{P}^\alpha < \mathcal{D}$. Comparing $\mathcal{B}$ with these ordered thresholds proves the three regions and both boundaries. If $Y_1/Y_0 > 1$, then $R_1/R_0 = (Y_1/Y_0)/\mathcal{P}^\alpha > 1$, excluding the fourth sign pattern. Finally, $\mathcal{B} = e^{\varepsilon(z_1-z_0)} \geq 1$. Holding the other ratios fixed and allowing $\varepsilon \geq 0$, a $\mathcal{B}$ in the joint-loss interval exists exactly when $\mathcal{D}\mathcal{P}^\alpha > 1$. □

Proof of Corollary 1. For $G = 0$, $Y = B/D$, $D' = -\alpha\omega'$, and

$$\frac{d \ln Y}{dz} = \varepsilon - \frac{D'}{D} = \varepsilon - \Lambda(s, \kappa), \quad \frac{d \ln R}{dz} = \frac{d \ln Y}{dz} + \alpha s. \quad (\text{A.13})$$

Let $d_A = 1 - \beta - \alpha\rho_A \geq 1 - \alpha - \beta > 0$. Since

$$D = d_A + \alpha s \Delta(\kappa), \quad \Delta_\kappa = -\delta_\rho, \quad D_\kappa = -\alpha s \delta_\rho, \quad (\text{A.14})$$

direct differentiation gives (22) and (23). From $\mathcal{M}_G = \omega/D$, $\omega_\kappa = s\delta_\rho$, and $D + \alpha\omega = 1 - \beta$, we obtain (24).

Write $h = (\eta - 1)(1 - s) > 0$. Then

$$\Lambda(s, \kappa) = \frac{\alpha s h \Delta(\kappa)}{d_A + \alpha s \Delta(\kappa)}. \quad (\text{A.15})$$

Equation (23) makes this function continuous and strictly decreasing in κ . Hence the endpoint condition around any q implies a unique solution in $(0, 1)$. Moreover, $\Lambda(s, 0) > q \geq 0$ implies $q < h$, and solving $\Lambda = q$ gives

$$\Delta(\kappa) = \frac{q d_A}{\alpha s (h - q)}.$$

Using $\Delta(\kappa) = \rho_A - \rho_P - \kappa\delta_\rho$ gives (25).

The function $\kappa(q)$ is strictly decreasing in q . Therefore the combined endpoint condition gives $\kappa_R^* < \kappa_Y^*$. Equation (21) and strict monotonicity of Λ give all three sign regions and the equality boundaries. At κ_R^* , real income is locally constant while nominal income falls; at κ_Y^* , nominal income is locally constant while real income rises. □

Proof of Corollary 2. Fix κ with $\Delta(\kappa) > 0$, and write

$$d_A = 1 - \beta - \alpha\rho_A > 0, \quad b = \alpha\Delta(\kappa) > 0, \quad C = \alpha(\eta - 1)\Delta(\kappa) > 0.$$

Because $D = d_A + bs$, direct differentiation gives

$$\Lambda_s(s, \kappa) = C \frac{d_A(1 - 2s) - bs^2}{(d_A + bs)^2}, \quad (\text{A.16})$$

$$\Lambda_{ss}(s, \kappa) = -\frac{2C d_A(d_A + b)}{(d_A + bs)^3} < 0. \quad (\text{A.17})$$

Thus Λ is strictly concave. The numerator $H(s) = d_A(1 - 2s) - bs^2$ of (A.16) satisfies $H(0) > 0$, $H(1) < 0$, and $H'(s) = -2d_A - 2bs < 0$. It therefore has one zero in $(0, 1)$. Solving $H(s) = 0$ yields (26); the strict inequality follows because $\sqrt{1 + b/d_A} > 1$.

Let $t = \sqrt{1 + b/d_A} > 1$. At $s^\dagger = 1/(1 + t)$, use $b = d_A(t^2 - 1)$ to obtain (27). Finally, $\Delta_\kappa = -\delta_\rho$ implies

$$\frac{\partial s^\dagger}{\partial \kappa} = \frac{\alpha\delta_\rho}{2d_A t(1 + t)^2} > 0, \quad (\text{A.18})$$

$$\frac{\partial \Lambda_{\max}}{\partial \kappa} = -\frac{(\eta - 1)\alpha\delta_\rho}{d_A t(1 + t)^2} < 0. \quad (\text{A.19})$$

These derivatives apply while $\Delta(\kappa) > 0$. □

A.1 Boundary and limiting cases

For finite z , Lemma 1 rules out channel corners. Instead, $z \rightarrow -\infty$ implies $s \rightarrow 0$ and $\omega \rightarrow \rho_A$, while $z \rightarrow \infty$ implies $s \rightarrow 1$ and $\omega \rightarrow \rho_P(\kappa)$. Likewise, $a \rightarrow 0$ and $a \rightarrow 1$ produce the respective degenerate-share limits, outside Assumption 1. Because $s \in (0, 1)$, $\omega = 0$ exactly when $\rho_A = \rho_P(\kappa) = 0$, giving $\mathcal{M}_G = 0$ and $Y = B/(1 - \beta)$. Likewise, $\omega = 1$ exactly when $\rho_A = \rho_P(\kappa) = 1$, giving $\mathcal{M}_G = m_0$.

The endpoints $\kappa = 0, 1$ produce $\rho_P(0) = \underline{\rho}_P$ and $\rho_P(1) = \bar{\rho}_P$, respectively. In the excluded limit $\delta_\rho = 0$, organizational capacity has no effect. If $\rho_A = \rho_P(\kappa)$, retention is locally invariant to z . If $\rho_A < \rho_P(\kappa)$, then $\omega_z > 0$; with $G = 0$ and $\varepsilon \geq 0$, nominal and real income rise. As $\alpha \rightarrow 0$,

$$Y \rightarrow \frac{B + \omega G}{1 - \beta}, \quad \mathcal{M}_G \rightarrow \frac{\omega}{1 - \beta};$$

as $\beta \rightarrow 0$, $D \rightarrow 1 - \alpha\omega$ and $\mathcal{M}_G \rightarrow \omega/(1 - \alpha\omega)$. Setting $\varepsilon = 0$ isolates retention and price effects. If $\alpha + \beta \uparrow 1$, contraction ceases to be uniform; at equality a finite fixed point remains only when $\omega < 1$, while $D = 0$ at full retention.

As $\eta \rightarrow 1$,

$$P_M \rightarrow p_A^{1-a} p_P^a, \quad s \rightarrow a, \quad (\text{A.20})$$

the Cobb–Douglas channel limit. As $\eta \rightarrow \infty$ at fixed unequal prices, demand converges to the cheaper channel. At equal prices the perfect-substitutes problem has multiple cost-minimizing allocations, although the CES sequence selects $s = a$.

For Corollary 2, $\Delta(\kappa) \downarrow 0$ implies $s^\dagger \uparrow 1/2$ and $\Lambda_{\max} \downarrow 0$. At $\Delta(\kappa) = 0$, $\Lambda \equiv 0$, so every share maximizes the zero function and the peak is not unique. If $\Delta(\kappa) < 0$, integration raises rather than lowers local retention; the drag-peak result does not apply.

Holding nominal $G > 0$ fixed, the general derivatives are

$$\frac{d \ln Y}{dz} = \frac{\varepsilon B + \omega_z G}{B + \omega G} + \frac{\alpha \omega_z}{1 - \beta - \alpha \omega}. \quad (\text{A.21})$$

$$\frac{d \ln R}{dz} = \frac{\varepsilon B + \omega_z G}{B + \omega G} + \frac{\alpha \omega_z}{1 - \beta - \alpha \omega} + \alpha s. \quad (\text{A.22})$$

For organizational capacity,

$$\frac{\partial \ln Y}{\partial \kappa} = s \delta_\rho \left[\frac{G}{B + \omega G} + \frac{\alpha}{D} \right] > 0, \quad \frac{\partial \ln R}{\partial \kappa} = \frac{\partial \ln Y}{\partial \kappa}. \quad (\text{A.23})$$

For a finite change define

$$\mathcal{N} = \frac{B_1 + \omega_1 G}{B_0 + \omega_0 G}. \quad (\text{A.24})$$

Then $Y_1/Y_0 = \mathcal{N}/\mathcal{D}$ and $R_1/R_0 = \mathcal{N}/(\mathcal{D}\mathcal{P}^\alpha)$; replacing $\mathcal{B}$ by $\mathcal{N}$ gives the general thresholds.